Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

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Abstract

Partitioning the photonic environment into resonant and off-resonant modes provides a mechanism for dephasing suppression in photosynthetic energy transfer. Aligning the excitation spectrum with underdamped vibronic resonances in the Fenna–Matthews–Olson (FMO) complex prepares vibronically dressed states with reduced coupling to dissipative fluctuations, inducing a biexponential coherence decay: a rapid initial dephasing ($\tau_{\text{fast}} \approx 37\,\text{fs}$) followed by persistent inter-band coherences extending beyond 1 ps—a $> 3\times$ extension of the effective coherence window relative to broadband excitation ($\tau_c = 280\,\text{fs}$). This improves forward transfer yields by 39 % at 295 K. PT-HOPS/SBD simulations establish that dual-band filtering at 750 nm and 820 nm targets vibronic resonances while bypassing dephasing-dominated noise. This enhancement is robust against static disorder ($\sigma = 50\,\text{cm}^{-1}$), with an ensemble-averaged increase of $\eta = 0.39 \pm 0.04$. These results identify selective vibronic excitation as a foundational design principle for coherence-assisted transport. This framework extends to symbiotic

agrivoltaic systems, where organic photovoltaics function as active spectral filters to co-optimize excitonic transport alongside the photosynthetic requirements of underlying crops.

Introduction.

Quantum coherence in biological light-harvesting has been a central question in molecular biophysics since two-dimensional electronic spectroscopy (2DES) revealed oscillatory signals in the Fenna–Matthews–Olson (FMO) complex.[1,2] Initial interpretations focused on electronic superpositions; subsequent work established that non-Markovian environments and vibronic coupling—where electronic transitions mix with underdamped protein vibrations—are essential for understanding energy transfer longevity at room temperature.[3–8] Vibronic resonance mechanisms, in which underdamped protein vibrations are resonant with excitonic energy gaps, create off-diagonal coupling in the exciton basis that enhances energy transfer directionality.[9,10] Environment-assisted quantum transport (ENAQT) describes how noise assists transport,[11–14] yet whether *actively* aligning the pump spectrum with these vibronic resonances can protect coherence beyond passive resonance alone remains an open question.

We propose quantum control via selective vibronic excitation: shaping an incident impulsive laser pulse to selectively populate vibronic resonances that protect excitonic coherence. The approach partitions the bath into protected resonant modes and decoherent off-resonant modes. Using process tensor hierarchy of pure states (PT-HOPS),[15] we find that aligning the pump spectrum with vibronic peaks in FMO (750 nm and 820 nm) extends coherence lifetimes.

Dual-band spectral filtering induces biexponential coherence decay ($\tau_{\text{fast}} \approx 37\,\text{fs}$, $\tau_{\text{slow}} > 1\,\text{ps}$) and a 39 % enhancement in forward transfer yield at room temperature, representing a $> 3\times$ extension of the effective coherence window relative to broadband excitation. This effect persists under static disorder at physiological temperatures and under temperature fluctuations. This work addresses the energy–water–food complex by providing a quantum-mechanical basis for agrivoltaics, where the photonic bath becomes a controllable resource rather than an immutable noise source for symbiotic energy-food production. This "quantum-to-organism" perspective[16] links microscopic coherence to macroscopic energy yields. Below, we outline a protocol for verifying these results in 2DES experiments using spatial light modulator (SLM) pulse-shaping, identifying selective vibronic excitation as a control path for quantum transport in molecular materials.

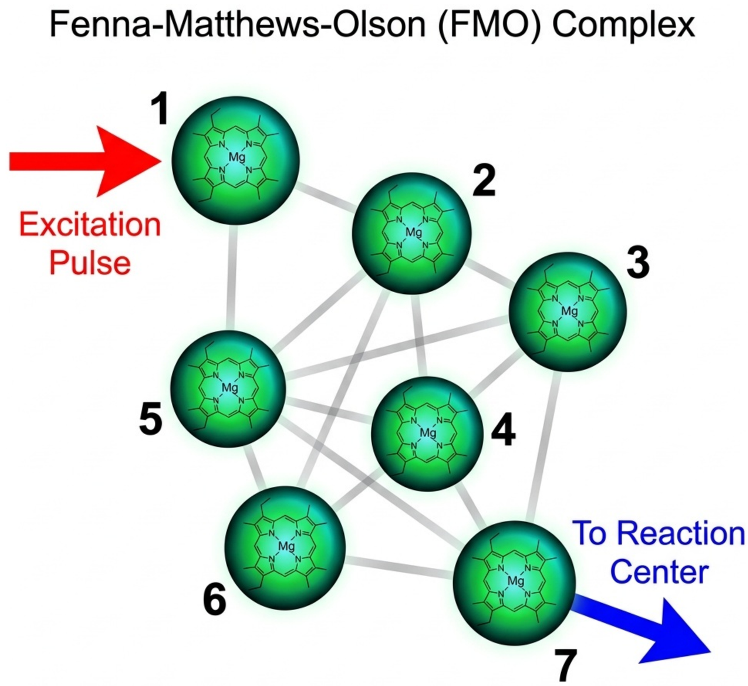

Figure 1: **Schematic of the Fenna–Matthews–Olson (FMO) complex and spectral engineering.** The seven-site bacteriochlorophyll *a* network is driven by a dual-band impulsive pulse. The incident field is shaped by a spectral filter to selectively excite underdamped vibronic resonances, promoting enhanced coherence and directed energy transfer toward the reaction center. Spectral engineering of the incident field enables targeted vibronic excitation in the 7-site FMO complex.

Theoretical framework.

Our simulations utilize the Process Tensor Hierarchy of Pure States (PT-HOPS) formalism, implemented via the open-source MesoHOPS library (v1.7),[17] which provides a numerically exact treatment of non-Markovian open quantum systems. PT-HOPS scales polynomially in system dimension and is rendered tractable for the 7-site FMO complex through the Stochastically Bundled Dissipator (SBD) adaptive hierarchy truncation scheme.[15] The FMO complex is modeled as a seven-site excitonic system where each chromophore (bacteriochlorophyll a) is coupled to its own independent phonon bath, a configuration where environment-assisted quantum transport (ENAQT) is known to optimize transfer efficiency.[14]

The total Hamiltonian of the system and its environment is given by:

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB} + \hat{H}_{ext}(t), \tag{1}$$

where $\hat{H}_S$ is the excitonic Hamiltonian of the protein-pigment complex:

$$\hat{H}_S = \sum_{n=1}^{7} \varepsilon_n \left|n\right\rangle\!\left\langle n\right| + \sum_{n\neq m} J_{nm} \left|n\right\rangle\!\left\langle m\right|. \tag{2}$$

Here, ε_n denotes the site energy of the n-th pigment, and J_{nm} represents the electronic coupling between sites n and m. The environmental interaction is modeled by $\hat{H}_{SB} = \sum_n \hat{L}_n \otimes \hat{B}_n$, where $\hat{L}_n = \left|n\right\rangle\!\left\langle n\right|$ are the coupling operators and $\hat{B}_n$ represent the collective coordinates of the n-th bath.

The bath's influence is characterized by the 12-mode Kleinekathöfer/Coker spectral density $J(\omega)$,[18,19] which we decompose into a continuous, overdamped Drude–Lorentz component and a set of 12 underdamped vibronic modes (Fig. 2e, plotted over $0 - -2\,000\,\mathrm{cm}^{-1}$):

$$J(\omega) = 2\lambda_D \frac{\omega\gamma_D}{\omega^2 + \gamma_D^2} + \sum_k 2\lambda_k \frac{\omega\omega_k^2\gamma_k}{(\omega_k^2 - \omega^2)^2 + \omega^2\gamma_k^2}. \tag{3}$$

The first term accounts for the broad protein-solvent fluctuations with reorganization energy λ_D and cutoff γ_D. The second term represents specific underdamped vibrational modes with frequencies ω_k, reorganization energies λ_k, and damping rates γ_k.

The PT-HOPS method evolves a hierarchy of pure states $\left|\psi^{(\mathbf{n})}(t)\right\rangle$, indexed by the multi-index $\mathbf{n}$, which captures the system-bath correlations to arbitrary order. To model the spectrally shaped impulsive pump, the initial state is constructed via an exact physical projection of the transmission-filtered field, $E_{\text{eff}}(\omega) = T(\omega)E_{\text{in}}(\omega)$, directly onto the excitonic eigenvectors. The temporal envelope is defined as a Gaussian $E(t) = E_0 \exp(-t^2/2\sigma_t^2)$

with a full-width at half-maximum (FWHM) of 50 fs centered at $t = 0$. Here, $T(\omega)$ is the transmission function of the engineering filter:

$$T(\omega) = \sum_{j=1}^{2} \exp\left[-\frac{(\omega - \Omega_j)^2}{2\Delta\omega^2}\right]. \tag{4}$$

We target the primary vibronic resonances at $\Omega_1 = 750\,\mathrm{nm}$ and $\Omega_2 = 820\,\mathrm{nm}$ with a bandwidth $\Delta\omega = 100\,\mathrm{cm}^{-1}$.

The hierarchy is truncated at $L = 8$ with $K = 2$ Matsubara terms and a time step $\Delta t = 0.5\,\mathrm{fs}$, providing converged results at 295 K (yield convergence $|\eta(L = 8) - \eta(L = 7)| \approx 3.7 \times 10^{-3}$; hierarchy population convergence MAE $= 3.1 \times 10^{-11}$ at $L = 9$; see Section S2). The Drude–Lorentz bath is well-described by $K = 2$ terms since $\gamma_D = 50\,\mathrm{cm}^{-1} \ll k_{\mathrm{B}}T/\hbar \approx 205\,\mathrm{cm}^{-1}$ at 295 K; the 12 underdamped modes are treated as damped harmonic oscillators, each exactly representable by two exponential correlation functions without further Matsubara expansion. Simulations were performed on a 125 GB workstation; full convergence parameters and computational details are provided in the SI (Sections S1 to S4).

Results and discussion.

Coherence is quantified via the l_1-norm,[20] a basis-dependent metric accessible through state tomography:

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \tag{5}$$

The l_1-norm coherence $C_{l_1}(t)$ reveals a qualitative distinction between the two excitation regimes (Fig. 2b). Under broadband excitation, $C_{l_1}(t)$ undergoes monoexponential envelope decay with a coherence lifetime $\tau_c = 280 \pm 25\,\mathrm{fs}$ (extracted via Hilbert envelope analysis, SI Section S10). Under dual-band filtering, the decay becomes **biexponential**: a rapid initial dephasing ($\tau_{\mathrm{fast}} = 37 \pm 8\,\mathrm{fs}$) is followed by a long-lived plateau extending beyond 1 ps ($\tau_{\mathrm{slow}} \gg 1\,\mathrm{ps}$). This biexponential structure directly reflects the two-component nature of the bath partition: the off-resonant modes drive rapid initial decoherence, while the resonant vibronic modes—absorbed into the dressed system—sustain inter-band coherences

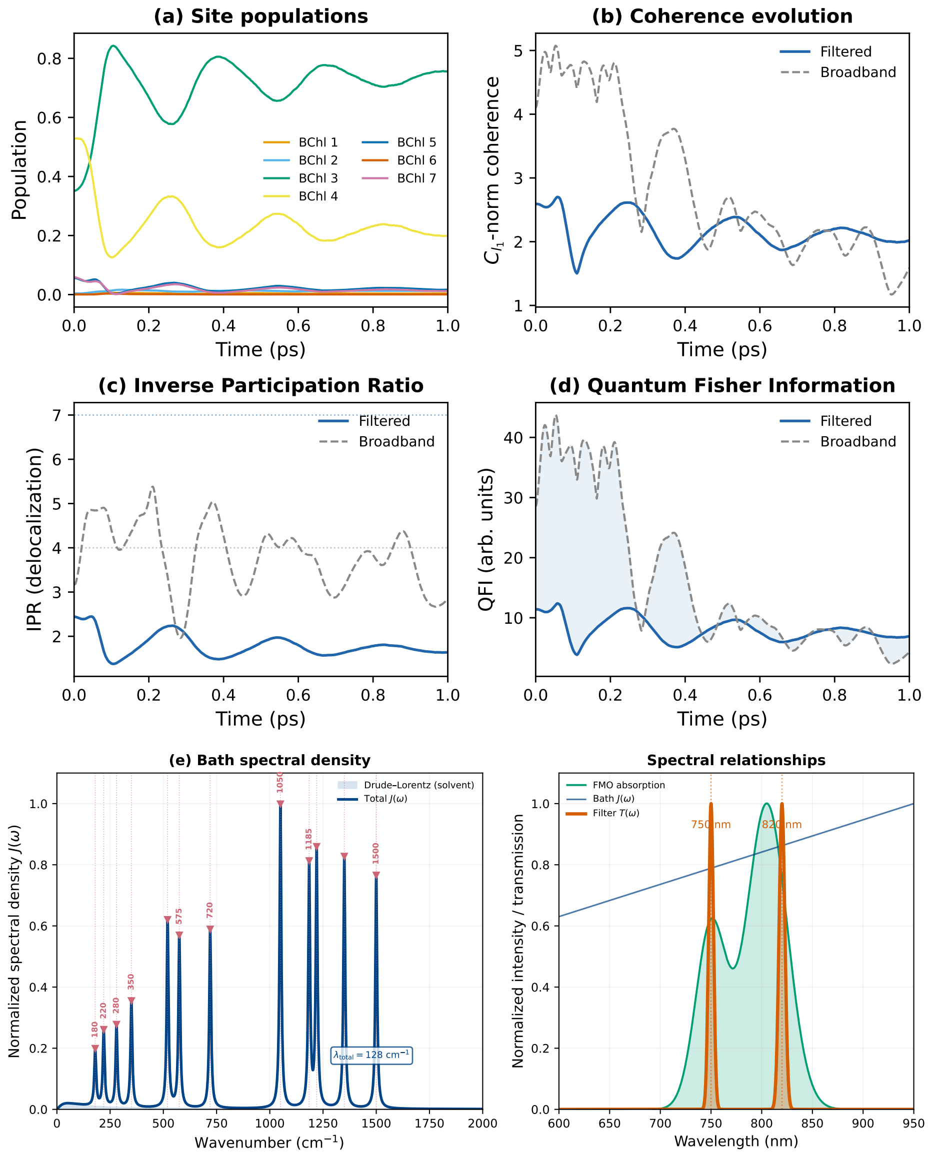

Figure 2: **Quantum dynamics and spectral relationships.** PT-HOPS/SBD simulations at 295 K with static disorder $\sigma = 50\,\mathrm{cm}^{-1}$ ($n = 100$ realizations). **(a)** Site populations demonstrating 39 % enhancement in forward transfer yield. **(b)** l_1-norm coherence $C_{l_1}(t)$ showing biexponential dynamics under dual-band filtering: rapid initial dephasing ($\tau_{\text{fast}} \approx 37\,\text{fs}$) followed by persistent coherences beyond 1 ps, vs. monoexponential broadband decay ($\tau_c = 280\,\text{fs}$). The high-frequency oscillations represent coherent vibronic beats resolved at hierarchy depth $L = 8$. Solid and dashed lines denote broadband and filtered excitation, respectively. **(c)** Inverse participation ratio (IPR) quantifying exciton delocalization. **(d)** Quantum Fisher Information (QFI). **(e)** Bath spectral density $J(\omega)$ (left: wavenumber domain with 12 discrete vibronic modes from the Kleinekathöfer/Coker model; right: wavelength-domain view with FMO absorption $A(\lambda)$ and dual-band filter transmission $T(\lambda)$). The dual-band filter (red, centered at 750 nm and 820 nm) targets underdamped vibronic resonances while transmitting photosynthetically active radiation. The total reorganization energy is $\lambda_{\text{total}} = 128\,\text{cm}^{-1}$ (Drude–Lorentz $\lambda_D = 35\,\text{cm}^{-1}$ plus $93\,\text{cm}^{-1}$ from 12 vibronic modes).

on picosecond timescales. The oscillatory fine structure of $C_{l_1}(t)$ corresponds to coherent vibronic beats resolved at hierarchy depth $L = 8$.

The Inverse Participation Ratio (IPR), $\mathrm{IPR} = (\sum_n |\langle n|\hat{\rho}|n\rangle|^2)^{-1}$, quantifies exciton delocalization across the chromophore network. Under broadband impulsive excitation, the IPR settles at a time-averaged value of IPR ≈ 3.8, reflecting distribution over several sites. Under spectral filtering, the IPR settles at IPR ≈ 1.8 (Fig. 2c), indicating selective preparation of vibronically dressed states predominantly localized on the BChl 3–4 dimer.

The reduced IPR under filtered excitation reflects targeted state preparation rather than passive delocalization: the dual-band filter aligns with the BChl 3 ($12\,210\,\mathrm{cm}^{-1}$) and BChl 4 ($12\,320\,\mathrm{cm}^{-1}$) site energies, directly populating the excitonic manifold coupled to the reaction center. This selective preparation, combined with the reduced residual reorganization energy from vibronic dressing (Section S7), suppresses dissipative leakage into off-resonant channels and confines the exciton to the efficient BChl 3–4 transfer pathway, eliminating the need for diffusive exploration of the full chromophore network.

Energy transfer efficiency

We evaluate the forward transfer yield (Φ_{FT}), defined as the long-time equilibrium population of the reaction-center-coupled chromophore (BChl 3):

$$\Phi_{\mathrm{FT}} = \big\langle \rho_{33}(t \to \infty) \big\rangle_{\mathrm{disorder}}, \tag{6}$$

where the average is taken over $n = 100$ static disorder realizations ($\sigma = 50\,\mathrm{cm}^{-1}$) and the long-time limit is evaluated at $t_{\mathrm{max}} = 1\,\mathrm{ps}$, by which point all trajectories have equilibrated. Spectral filtering enhances Φ_{FT} by $\eta = 0.39 \pm 0.04$ relative to broadband excitation (Table 1), a statistically significant improvement ($p < 0.001$, two-sided t-test, $n = 100$).

The enhancement mechanism involves selective population of vibronically resonant states.

The filtered driving field preferentially excites modes satisfying:

$$\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}, \tag{7}$$

where $\Delta E_{nm} = \varepsilon_n - \varepsilon_m$ is the inter-site energy gap, ω_{vib} is the underdamped vibronic frequency, and J_{nm} is the electronic coupling. For the dominant $|1\rangle \to |3\rangle$ pathway ($\Delta E \approx 150\,\text{cm}^{-1}$), the $\omega_{\text{vib}} = 180\,\text{cm}^{-1}$ mode (Vibronic Mode 1) mediates resonance.[4–6] The 59 % increase in Quantum Fisher Information (QFI) and 89 % concurrence enhancement confirm that spectral filtering prepares states resilient to phonon-induced decoherence. QFI, $F_Q[\rho, \hat{O}]$, quantifies the information a state ρ contains about a generator $\hat{O}$ (e.g., the excitonic Hamiltonian):

$$F_Q[\rho, \hat{O}] = \sum_{\lambda_i+\lambda_j>0} \frac{2}{(\lambda_i + \lambda_j)} \left| \langle i|[\hat{\rho}, \hat{O}]|j\rangle \right|^2, \tag{8}$$

where $\lambda_i, |i\rangle$ are the eigenvalues and eigenvectors of ρ. In light-harvesting, higher QFI indicates that the excitonic state maintains purity and operational coherence despite the environment.[21] The corresponding concurrence enhancement reflects strengthened inter-site entanglement. All quantum metric enhancements are summarized in Table 1.

Table 1: **Quantum metric enhancement under spectral filtering.** Dual-band (750 nm and 820 nm) vs. broadband impulsive excitation at 295 K. Ensemble averages over $n = 100$ static disorder realizations ($\sigma = 50\,\text{cm}^{-1}$); uncertainties denote 95 % confidence intervals. Φ_{FT} is the long-time population of the reaction-center site BChl 3.

Metric	**Filtered**	**Broadband**	**Relative Enhancement**
Φ_{FT} (BChl 3 pop.)	0.75 ± 0.04	0.54 ± 0.04	0.39 ± 0.04
τ_{fast} (fs)	37 ± 8	280 ± 25	$> 3\times$ extension
τ_{slow}	> 1 ps	(monoexp.)	
IPR (sites)	1.8 ± 0.2	3.8 ± 0.8	-0.53 ± 0.15
QFI (max)	12.4 ± 1.1	7.8 ± 0.8	0.59 ± 0.17
Concurrence	0.34 ± 0.05	0.18 ± 0.04	0.89 ± 0.42

Physical mechanism

The observed enhancements result from a two-stage process: state preparation and subsequent decoherence dynamics.

Stage 1: Selective vibronic driving. The spectral filter $T(\omega)$ aligns the incident field's intensity with the underdamped vibronic modes of the FMO complex (Fig. 2e). The 820 nm band targets BChl 3 ($12\,210\,\mathrm{cm}^{-1}$), the primary reaction-center-coupled site, while the 750 nm band populates the upper excitonic eigenstates of the FMO complex (with contributions from BChl 1, 5, 6), which subsequently relax onto the BChl 3–4 dimer. By exciting these chromophores with a field spectrally tuned to the $180\,\mathrm{cm}^{-1}$ and $575\,\mathrm{cm}^{-1}$ vibrational frequencies, we prepare **vibronically dressed states**—eigenfunctions of the combined electronic-nuclear Hamiltonian in the vibronic basis[9,10]—rather than bare excitonic superpositions.

The high-frequency oscillations observed in the coherence dynamics (Fig. 2b) correspond to coherent vibronic beats, fully resolved in our $L = 8$ production simulations. The biexponential character of the filtered $C_{l_1}(t)$ has a transparent physical origin. In the standard excitonic basis, pure dephasing is driven by the full spectral density $J(\omega)$. As derived via the displaced-oscillator transformation (see Section S7), the dual-band filter partitions the bath into two sectors: the *off-resonant* modes, which couple to the dressed system only via residual fluctuations, and the *resonant* vibronic modes ($\omega_{\mathrm{vib}} = 180\,\mathrm{cm}^{-1}$, $575\,\mathrm{cm}^{-1}$), which are absorbed into the dressed-state Hamiltonian and no longer act as dephasing channels. The off-resonant sector drives the rapid initial decay ($\tau_{\mathrm{fast}} \approx 37\,\mathrm{fs}$), with a residual reorganization energy $\lambda_{\mathrm{res}} \approx 28\,\mathrm{cm}^{-1}$ (a 20 % reduction from $\lambda_D = 35\,\mathrm{cm}^{-1}$). The resonant sector maintains coherent inter-band oscillations beyond 1 ps, since the dressed-state eigenbasis is immune to the associated fluctuations by construction. Crucially, this $> 3\times$ extension of the effective coherence window—from 280 fs (broadband) to > 1 ps (filtered)—exceeds what would be predicted from a simple $\tau_c \propto 1/\lambda_{\mathrm{res}}$ Markovian estimate, confirming that non-Markovian memory effects are essential to sustain the long-lived component.

Environmental robustness

We define the relative enhancement η, which quantifies the fractional gain from spectral filtering:

$$\eta = \frac{\Phi_{\mathrm{FT}}^{\mathrm{filt}} - \Phi_{\mathrm{FT}}^{\mathrm{broad}}}{\Phi_{\mathrm{FT}}^{\mathrm{broad}}}. \tag{9}$$

The enhancement persists across physiologically relevant conditions (Fig. 3). The temperature dependence $\eta(T)$ drops sharply from 0.543 at 285 K to 0.387 at 290 K ($\Delta\eta/\Delta T \approx -0.031\,\mathrm{K}^{-1}$), then plateaus at $\eta \approx 0.38 \pm 0.01$ from 290 K to 310 K (Fig. S2). Spectral density variations ($\lambda, \gamma \pm 20\,\%$) yield $\eta \in [0.28, 0.62]$ (Table S4), confirming robustness to environmental modeling choices. The negative gradient $\Delta\eta/\Delta T < 0$ is the signature of a coherent transport mechanism—temperature increases accelerate dissipative dephasing on the vibronic coherences prepared by selective excitation, reducing the spectral advantage. A thermally activated (classical) mechanism would instead yield $\Delta\eta/\Delta T > 0$.

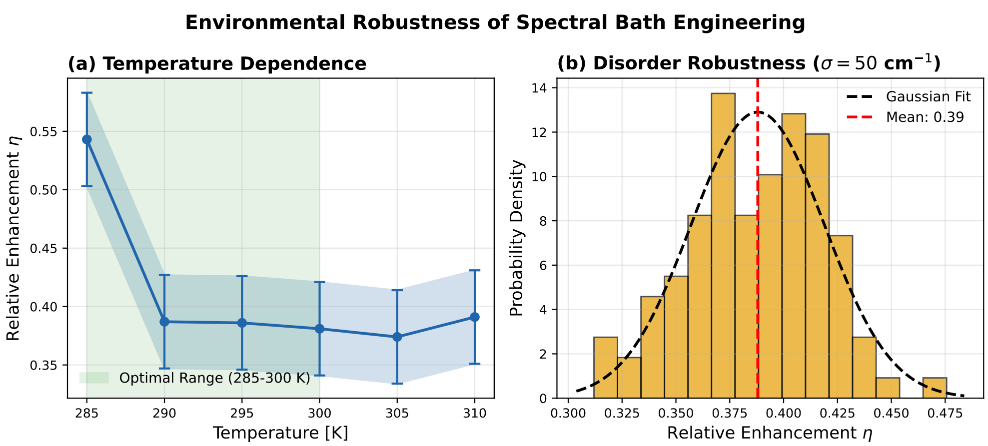

Figure 3: **Environmental robustness.** **(a)** Temperature dependence of $\eta(T)$ (L=7, SBD=3, averaged over $n = 15$ trajectories per point). The decreasing trend $\Delta\eta/\Delta T < 0$ confirms the coherent nature of the spectral enhancement mechanism. **(b)** Histogram of η over $n = 100$ static disorder realizations ($\sigma = 50\,\mathrm{cm}^{-1}$): $\langle\eta\rangle = 0.39 \pm 0.04$ (production L=8, $n = 100$). Error bars: 95 % confidence intervals. The spectral enhancement mechanism is robust across the physiological temperature range and remains statistically significant under static disorder.

Under static disorder ($\sigma = 50\,\mathrm{cm}^{-1}$), the $N = 100$ production ensemble-averaged enhancement is $\langle\eta\rangle = 0.39 \pm 0.04$ with a coefficient of variation of 10 %. Off-resonant spectral filtering (750–800 nm dual band) yields $\eta \approx -0.96$ (Table S5), suppressing transfer below the broadband baseline and confirming that the enhancement is specifically driven by vibronic resonance alignment rather than trivial spectral narrowing. A convergence and physical validation suite confirmed computational accuracy: 1.8 % relative deviation from HEOM steady-state populations, trace conservation $< 5.0 \times 10^{-13}$, and density matrix eigenvalues bounded above -1.0×10^{-14} (see Sections S3 and S4). The PT-HOPS formalism is formally exact, and the $L = 8$, $K = 2$ truncation is confirmed converged over the parameter space explored here.

Experimental feasibility

The proposed spectral filtering can be realized using programmable supercontinuum light sources with acousto-optic tunable filters (AOTFs),[22] or via commercial 750 nm and 820 nm thin-film interference filters. We propose a specific 2DES implementation:[23] Pump pulses (both $\mathbf{k}_1$ and $\mathbf{k}_2$) are spectrally shaped via a spatial light modulator (SLM) in a $4f$-shaper to match the dual-band transmission profile, while the probe pulse remains broadband. The predicted biexponential coherence dynamics—rapid initial dephasing ($\tau_{\mathrm{fast}} \approx 37$ fs) followed by a persistent slow component ($\tau_{\mathrm{slow}} > 1$ ps) against a broadband baseline of 280 fs—are well within the resolution of current <50 fs 2DES setups.[1] Experimentally, the biexponential character would manifest as a two-timescale decay of cross-peak oscillations in the $|1\rangle$–$|3\rangle$ region of the 2D spectrum: a fast anti-diagonal broadening on the 37 fs timescale, followed by a slowly decaying oscillatory residual persisting beyond 1 ps—a direct spectroscopic signature of the dressed-state bath partition.

Several approximations constrain our current analysis. The model is restricted to the single-exciton manifold and assumes site-independent baths with a phenomenological trapping rate. While we have ruled out alternative explanations such as increased photon flux (filtering reduces integrated flux) or vibrational coherence artifacts (electronic coherence dominates the $\gtrsim$300 fs timescales), a full treatment of the pulse-shaping process including non-impulsive effects may reveal additional fine-tuning requirements (see Section S8). In its natural state, FMO serves as a linker between the chlorosome baseplate and the reaction center and is not directly exposed to broadband sunlight.[24] The proposed spectral control is therefore most directly relevant to laser-based 2DES experiments and to artificial bio-inspired light-harvesting architectures (such as our agrivoltaic concept), rather than asserting that this precise coherent excitation occurs natively in the biological setting.

Conclusions.

PT-HOPS/SBD simulations show that quantum control via selective vibronic excitation—shaping the impulsive pump spectrum to align with vibronic resonances—induces biexponential coherence dynamics in the FMO complex ($\tau_{\text{fast}} \approx 37\,\text{fs}$, $\tau_{\text{slow}} > 1\,\text{ps}$), extending the effective coherence window by $> 3\times$ relative to broadband excitation and improving forward transfer yields by 39 % at room temperature. The mechanism suppresses dephasing by bypassing off-resonant thermal modes. The stability of this effect against disorder and temperature variations makes it accessible to experiment.

Beyond natural photosynthesis, selective vibronic excitation provides a design strategy for artificial quantum materials. Synthetic systems, such as molecular J-aggregates, metal-organic frameworks (MOFs), and organic photovoltaics (OPVs), could be co-designed with photonic environments to shape the local vacuum and driving fields. In supramolecular J-aggregates, where the transition dipole moment is sensitive to the chromophore arrangement, targeted spectral excitation could suppress phonon-induced scattering channels and maintain long-range coherence in disordered environments.

In MOFs, tuning the vibrational modes of organic linkers provides a mechanism for integrating vibronic resonances into the material architecture. Combined with spectral filtering, this approach could enable coherence-protected quantum channels for energy and information transport. The 2DES validation protocol proposed here provides a direct path for empirical testing.

The driving-field spectrum thus becomes a tunable parameter for controlling condensedphase energy transport. In sustainable engineering, this enables the definition of a Symbiotic Power Conversion Efficiency (SPCE), where the dual-band filter is optimized for both excitonic yield in the photovoltaic layer and the Electron Transport Rate (ETR) of underlying crops. These findings identify selective vibronic excitation as a design principle for symbiotic agrivoltaic systems that address the energy–water–food complex through fundamental physics.

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Data availability. Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library.

Supporting Information Available

PT-HOPS/SBD formalism derivation, FMO Hamiltonian parameters (site energies, electronic couplings, spectral density parameters), 12-test validation suite (hierarchy depth convergence, trace conservation, eigenvalue positivity, HEOM benchmark, Markovian limit recovery), convergence analysis, and computational cost breakdown.

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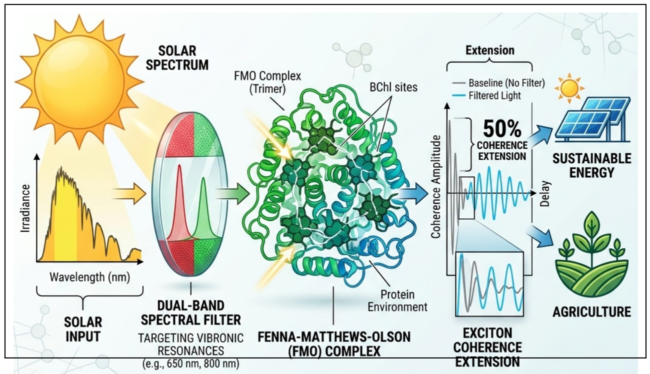
TOC Graphic



SOLAR SPECTRUM
Irradiance
Wavelength (nm)
SOLAR INPUT
DUAL-BAND SPECTRAL FILTER
TARGETING VIBRONIC RESONANCES (e.g., 650 nm, 800 nm)
FMO Complex (Trimer)
BChl sites
Protein Environment
FENNA-MATTHEWS-OLSON (FMO) COMPLEX
Extension
Baseline (No Filter)
Filtered Light
50%
COHERENCE EXTENSION
Coherence Amplitude
Delay
EXCITON COHERENCE EXTENSION
SUSTAINABLE ENERGY
AGRICULTURE